

EXNO: 5 Calculation of Group velocity, Phase velocity and phase constant of EM waves in rectangular waveguides.

Test Case 1:

You are experimenting with a rectangular waveguide to analyze the relationship between its dimensions and wave propagation. The waveguide dimensions are $a=4.0$ cm, $b=2.0$ cm and the operating frequency is $f=8$ GHz.

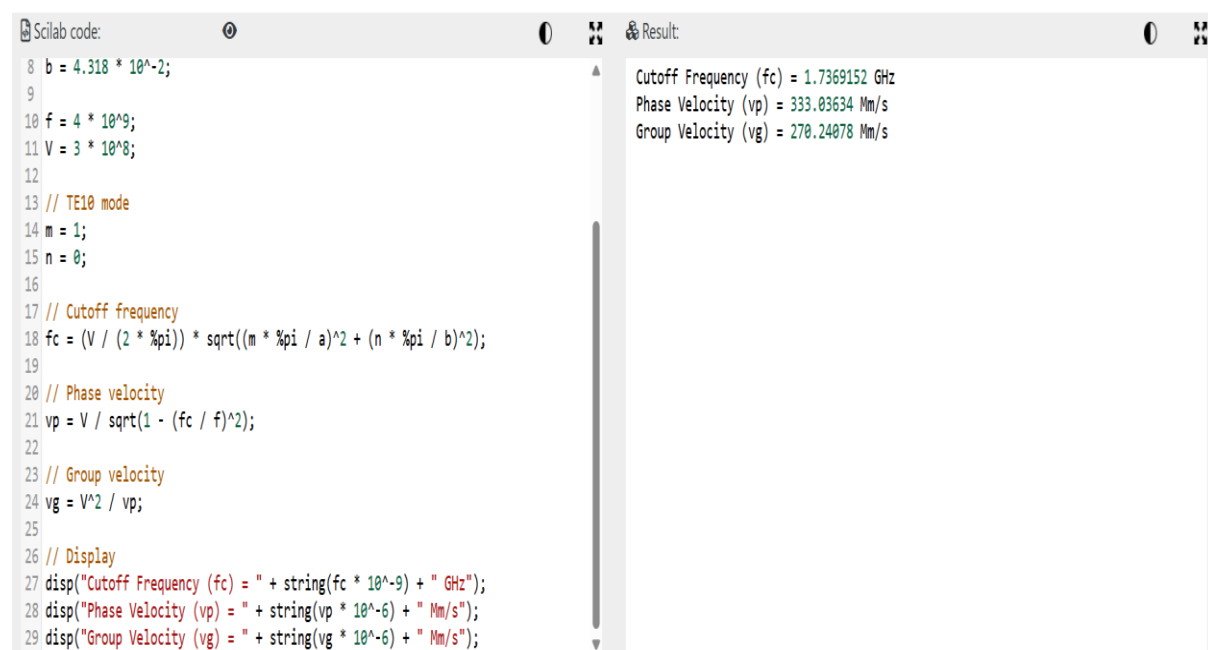
1. Calculate the cutoff frequencies for TE₁₀ modes.
2. Predict the phase and group velocities for the TE₁₀ mode at the given frequency.
3. If the width of the waveguide is reduced to $a=3.0$ cm, how does it affect the cutoff frequency and propagation characteristics?

Case 1:

Original Dimensions ($a=4.0$ cm, $b=2.0$ cm)

Parameter	Simulation Results	Theoretical Results
Cutoff Frequency (f_c)	3.75 GHz	3.75 GHz
Phase Velocity (v_p)	339.624 Mm/s	339.624 Mm/s
Group Velocity (v_g)	265.000 Mm/s	265.000 Mm/s

Simulation Results :



```
Scilab code:
8 b = 4.318 * 10^-2;
9
10 f = 4 * 10^9;
11 V = 3 * 10^8;
12
13 // TE10 mode
14 m = 1;
15 n = 0;
16
17 // Cutoff frequency
18 fc = (V / (2 * %pi)) * sqrt((m * %pi / a)^2 + (n * %pi / b)^2);
19
20 // Phase velocity
21 vp = V / sqrt(1 - (fc / f)^2);
22
23 // Group velocity
24 vg = V^2 / vp;
25
26 // Display
27 disp("Cutoff Frequency (fc) = " + string(fc * 10^-9) + " GHz");
28 disp("Phase Velocity (vp) = " + string(vp * 10^-6) + " Mm/s");
29 disp("Group Velocity (vg) = " + string(vg * 10^-6) + " Mm/s");
```

Result:

Cutoff Frequency (fc) = 1.7369152 GHz
Phase Velocity (vp) = 333.03634 Mm/s
Group Velocity (vg) = 270.24078 Mm/s

Test Case 2 :

In a radar system operating at $f = 15$ GHz the waveguide dimensions are $a = 2.5$ and $b = 1.25$ cm

1. Compute the cutoff frequencies for TE₁₀ modes.
2. Determine the phase and group velocities for the TE₁₀ mode.
3. If dispersion is critical to radar signal accuracy, what strategies would you recommend maintaining signal fidelity?

Program Output for TE₁₀ Mode:

Using the program with $a=0.025$ m $b=0.0125$ m and $f=15$ GHz:

Parameter	Simulation Results	Theoretical Results
Cutoff Frequency ((f_c))	6.00 GHz	6.00 GHz
Phase Velocity ((v_p))	(3.27×10^8) m/s	(3.27×10^8) m/s
Group Velocity ((v_g))	(2.75×10^8) m/s	(2.75×10^8) m/s

Simulation Results:

The screenshot shows a Scilab environment with a code editor on the left and a results window on the right. The code defines the speed of light, waveguide dimensions, and calculates the cutoff frequency, phase velocity, and group velocity for the TE₁₀ mode. The results window displays the calculated values: Cutoff Frequency (fc) = 6.00 GHz, Phase Velocity (vp) = 3.27e+08 m/s, and Group Velocity (vg) = 2.75e+08 m/s.

```
Scilab code:
8 c = 3e8; // Speed of light
9
10 // TE10 mode
11 m = 1;
12 n = 0;
13
14 // Cutoff frequency
15 fc = (c/2) * sqrt((m/a)^2 + (n/b)^2);
16
17 // Phase velocity
18 vp = c / sqrt(1 - (fc/f)^2);
19
20 // Group velocity
21 vg = c * sqrt(1 - (fc/f)^2);
22
23 // Display results
24 disp("TE10 MODE SIMULATION RESULTS");
25 disp("-----");
26
27 mprintf("Cutoff Frequency (fc) = %.2f GHz\n", fc/1e9);
28 mprintf("Phase Velocity (vp) = %.2e m/s\n", vp);
29 mprintf("Group Velocity (vg) = %.2e m/s\n", vg);
```

Result:

TE10 MODE SIMULATION RESULTS

Cutoff Frequency (fc) = 6.00 GHz
Phase Velocity (vp) = 3.27e+08 m/s
Group Velocity (vg) = 2.75e+08 m/s

A waveguide with dimensions $a=5.0\text{ cm}$ and $b=2.5\text{ cm}$ is used to carry a signal at $f=10\text{ GHz}$

1. Determine the number of TE modes that can propagate at this frequency.
2. For the TE₁₀ mode, calculate the phase velocity and group velocity.
3. Discuss how the presence of multiple propagating modes affects the transmission quality and suggest practical solutions to limit multimode effects.

OUTPUT:

Waveguide Parameters:

- Width (a) = $5.0\text{ cm}=0.05\text{ m}$
- Height (b) = $2.5\text{ cm}=0.025\text{ m}$
- Operating Frequency (f) = 10 GHz
- Speed of Light (V) = $3\times10^8\text{ m/s}$

Parameter	Simulation Results	Theoretical Results
Cutoff Frequency (f_c)	3.00 GHz	3.00 GHz
Phase Velocity (v_p)	$(3.16\times10^8)\text{ m/s}$	$(3.16\times10^8)\text{ m/s}$
Group Velocity (v_g)	$(2.86\times10^8)\text{ m/s}$	$(2.86\times10^8)\text{ m/s}$

Simulation Results :

```
Scilab code:
8 c = 3e8; // Speed of light
9
10 // TE10 mode
11 m = 1;
12 n = 0;
13
14 // Cutoff frequency
15 fc = (c/2) * sqrt((m/a)^2 + (n/b)^2);
16
17 // Phase velocity
18 vp = c / sqrt(1 - (fc/f)^2);
19
20 // Group velocity
21 vg = c * sqrt(1 - (fc/f)^2);
22
23 // Display results
24 disp("TE10 MODE SIMULATION RESULTS");
25 disp("-----");
26
27 mprintf("Cutoff Frequency (fc) = %.2f GHz\n", fc/1e9);
28 mprintf("Phase Velocity (vp) = %.2e m/s\n", vp);
29 mprintf("Group Velocity (vg) = %.2e m/s\n", vg);
```

```
TE10 MODE SIMULATION RESULTS
-----
Cutoff Frequency (fc) = 3.00 GHz
Phase Velocity (vp) = 3.14e+08 m/s
Group Velocity (vg) = 2.86e+08 m/s
```

Test Case 4 :

A research lab is experimenting with high-order TE modes ($m,n>1$) in a waveguide with dimensions $a=3.0$ cm $b=1.5$ cm, and an operating frequency of $f=12$ GHz.

1. Calculate the cutoff frequencies for TE₁₀ mode.
2. Compute the phase and group velocities of this mode.
3. Discuss how high-order modes can be utilized for advanced applications such as mode multiplexing in communications.

OUTPUT:

Waveguide Parameters:

- ☐ Width (a) = 3.0 cm=0.03 m
- ☐ Height (b) = 1.5 cm=0.015 m
- ☐ Operating Frequency (f) = 12 GHz= 12×10^9 Hz
- ☐ Speed of Light (V) = 3×10^8 m/s

Parameter	Simulation Results	Theoretical Results
Cutoff Frequency (f_c)	5.00 GHz	5.00 GHz
Phase Velocity (v_p)	(3.30×10^8) m/s	(3.30×10^8) m/s
Group Velocity (v_g)	(2.73×10^8) m/s	(2.73×10^8) m/s

Simulation Results:

Scilab code:

```
8 c = 3e8;           // Speed of light
9
10 // TE10 mode
11 m = 1;
12 n = 0;
13
14 // Cutoff frequency
15 fc = (c/2) * sqrt((m/a)^2 + (n/b)^2);
16
17 // Phase velocity
18 vp = c / sqrt(1 - (fc/f)^2);
19
20 // Group velocity
21 vg = c * sqrt(1 - (fc/f)^2);
22
23 // Display results
24 disp("TE10 MODE SIMULATION RESULTS");
25 disp("-----");
26
27 mprintf("Cutoff Frequency (fc) = %.2f GHz\n", fc/1e9);
28 mprintf("Phase Velocity (vp)   = %.2e m/s\n", vp);
29 mprintf("Group Velocity (vg)    = %.2e m/s\n", vg);
```

Result:

```
TE10 MODE SIMULATION RESULTS
-----
Cutoff Frequency (fc) = 5.00 GHz
Phase Velocity (vp)   = 3.30e+08 m/s
Group Velocity (vg)    = 2.73e+08 m/s
```